

Dataset Features in the EPSS-TPG Experiments

1 Overview

The experiments use two families of datasets. The first is the Sec4AI4Aec-EPSS-Enhanced LLM-summary corpus: GPT, Gemma, Mistral and DeepSeek (refreshed in this round, with Mistral as a new addition), plus the older NVD/KEV labelled corpus where the binary label is membership in the CISA Known Exploited Vulnerabilities (KEV) catalog. The second family is the Megavul branch: GPT, Mistral and Gemma generated against the Megavul commit-level vulnerability corpus. Megavul rows are commit-grounded: each row carries the actual vulnerable code, the patched code, and the commit URL alongside the LLM summaries. The Megavul datasets share a common 29-column schema and a different label inventory from the social-media datasets (Section 2.5).

The pipeline does not consume every raw column directly. Each dataset is converted into a unified labelled-records form, and the training code then builds two kinds of input from those records:

- a Text Property Graph (TPG), built from the CVE description and, optionally, one or more LLM summary columns;
- a tabular feature vector, used only when the hybrid model is enabled.

For EPSS soft-label prediction the target is EPSS itself, so the EPSS score and percentile are removed from the tabular input via the `--no-epss-feature` flag. The NVD/KEV binary experiments target KEV membership instead, so EPSS can stay in the tabular branch without leaking the label.

2 Datasets

The four social-media LLM-summary datasets share most of their schema. They are not interchangeable: each dataset’s summary columns were generated by a different LLM, the missing-summary patterns differ by dataset, and DeepSeek uses an older column naming.

2.1 GPT Summary Dataset

GPT contains 9,218 records covering 9,218 unique CVEs (one row per CVE). Each row carries the NVD description, the CVSS components, EPSS score and status, social and GitHub source counts, public-code availability, and three GPT-generated summary columns.

2.2 Gemma Summary Dataset

Gemma has the same row count, the same CVE coverage and the same 28-column schema as GPT. The difference is the summary text: this dataset’s summaries are generated by Gemma.

Property	Value
Records	9,218
Columns	28
Unique CVEs	9,218
Target in soft-label runs	EPSS score
Main TPG text	CVE description
Summary columns available	all-sources, GitHub-URLs, CVSS-metrics
All-sources summary non-empty	7,606 (82.5%)
GitHub-URLs summary non-empty	2,246 (24.4%)
CVSS-metrics summary non-empty	9,218 (100%)
Records with public code	2,339

Table 1: GPT summary dataset.

Property	Value
Records	9,218
Columns	28
Unique CVEs	9,218
Target in soft-label runs	EPSS score
Main TPG text	CVE description
All-sources summary non-empty	7,636 (82.8%)
GitHub-URLs summary non-empty	2,381 (25.8%)
CVSS-metrics summary non-empty	9,218 (100%)
Records with public code	2,339

Table 2: Gemma summary dataset.

Gemma is the controlled counterpart to GPT: the structured fields are identical, only the LLM that produced the summaries is different. In baseline runs only the description is used. In summary runs the relevant summary column is mapped to the unified *LLM summary* field before the TPG is built.

2.3 Mistral Summary Dataset (New)

Mistral was added in this round. Same row count, same CVE coverage, same 28-column schema as GPT and Gemma, same three summary columns.

Property	Value
Records	9,218
Columns	28
Unique CVEs	9,218
Target in soft-label runs	EPSS score
Main TPG text	CVE description
All-sources summary non-empty	7,317 (79.4%)
GitHub-URLs summary non-empty	2,388 (25.9%)
CVSS-metrics summary non-empty	9,016 (97.8%)
Records with public code	2,339

Table 3: Mistral summary dataset.

2.4 DeepSeek Summary Dataset

DeepSeek has the same row count and CVE coverage but uses the older 27-column schema: the date / time / text / source-count / code-available / delta-days fields take the place of the newer date-posted / time-posted / social-media-post / occurrence-count / GitHub-links-with-code-available / days-since-git-source fields. It also has only two of the three summary columns: the all-sources and GitHub-URLs summaries are present, the CVSS-metrics summary is not. The dataset adapter handles either schema, so DeepSeek still works in the pipeline; the S_cvss variant is the one that cannot run on it.

Property	Value
Records	9,218
Columns	27 (older schema)
Unique CVEs	9,218
Target in soft-label runs	EPSS score
Main TPG text	CVE description
All-sources summary non-empty	7,636 (82.8%)
GitHub-URLs summary non-empty	2,381 (25.8%)
CVSS-metrics summary non-empty	column not present
Records with public code	2,339

Table 4: DeepSeek summary dataset. Older schema, no CVSS-metrics summary.

2.5 Megavul Datasets (New)

The Megavul datasets come from the Megavul branch of the same upstream repository. There are three datasets: Mega-GPT, Mega-Mistral and Mega-Gemma, each with the same 9,486 rows covering 9,240 unique CVEs. Rows are not one-per-CVE: a single CVE can appear in several rows when the upstream Megavul corpus links it to more than one vulnerable commit. The pipeline deduplicates by CVE-ID when building the labelled records.

The schema is different from the social-media datasets. There is no social-media post field, no observation date, no source-platform column. Instead, every row carries a commit URL, the pre-patch and post-patch function code, and CVSS information in two flavours: the original NVD CVSS, plus an LLM-augmented v3 estimate when the original is v2-only. The summary columns are also different: the three LLM summaries are a commit-URL summary, a code summary (an LLM summary of the vulnerable function), and a CVSS-metrics summary. There is no all-sources or GitHub-URLs summary.

2.5.1 Megavul Label Inventory

The Megavul datasets carry several columns that could in principle act as a training label. Their coverage and class balance are summarised below (percentages are over the raw 9,486 rows).

Two facts dominate the choice of training target.

- KEV membership only marks 37 Megavul rows (about 22 unique CVEs once duplicates are collapsed). After a 70/15/15 split that leaves roughly three positives in the test set, which is too sparse to estimate precision, recall or F1 with any confidence. KEV is not a viable training label on Megavul on its own.

Property	Mega-GPT	Mega-Mistral	Mega-Gemma
Raw rows	9,486	9,486	9,486
Unique CVEs	9,240	9,240	9,240
Columns	29	29	29
Source language: C/C++ rows	8,593	8,593	8,593
Source language: Java rows	893	893	893
Commit-URL summary non-empty	9,469 (99.8%)	9,486 (100%)	9,486 (100%)
Code summary non-empty	9,256 (97.6%)	9,486 (100%)	9,486 (100%)
CVSS-metrics summary non-empty	9,486 (100%)	9,305 (98.1%)	9,486 (100%)

Table 5: Megavul dataset sizes and summary-column coverage. The three datasets share the same row set; only the LLM that produced the summaries differs.

Candidate label column	Non-empty rows	Coverage	Comment
KEV date-added	37	0.39%	CISA KEV catalog membership; hard binary signal.
CVSS base severity	9,486	100%	4-class categorical: LOW / MEDIUM / HIGH / CRITICAL
CVSS v3-augmented severity	3,510	37.0%	v3 augmented severity, only for CVEs whose original NVD C
EPSS at publication	varies	partial	EPSS at CVE publication time; often missing.
EPSS most-recent	6,694	70.6%	EPSS at the most recent recalculation; the pipeline’s primary

Table 6: Label-candidate columns in the Megavul datasets.

- EPSS coverage is 70.6% when the most-recent EPSS value is preferred and the publication-time EPSS is used as a fallback. The remaining 29.4% of rows have no EPSS at all and are dropped by the dataset adapter.

The Megavul soft-label runs use the EPSS score as the continuous regression target, exactly like the social-media runs. The binary label used for stratified splitting and for threshold-based metrics is the same EPSS-thresholded value: positive iff $\text{EPSS} \geq 0.10$.

Positive/negative ratio at several thresholds. The EPSS distribution is heavily skewed on Megavul; the positive rate is about seven times sparser than on the social-media datasets. The numbers below are over the 9,486 rows of the Mega-GPT dataset; the other two Megavul datasets give the same numbers because EPSS is taken from the shared NVD/EPSS metadata, not from the LLM summaries.

EPSS binarisation threshold	Positives	Negatives	Positive rate	Comment
≥ 0.01	1,840	7,646	19.40%	Very lenient; includes most non-trivial scores.
≥ 0.05	280	9,206	2.95%	Moderate.
≥ 0.10	164	9,322	1.73%	Pipeline default.
≥ 0.20	124	9,362	1.31%	Tight.
≥ 0.50	94	9,392	0.99%	Very tight.

Table 7: Megavul EPSS positive/negative split at several binarisation thresholds. The pipeline default of 0.10 yields a $\sim 1.73\%$ positive rate.

The skew matters for training. With a $\sim 2\%$ positive rate, a naive classifier that always predicts “negative” already reaches 98% accuracy and the random-baseline PR-AUC equals the positive rate

(≈ 0.02). Reported metrics on Megavul should be read against that baseline, not against the much denser ($\sim 15.5\%$) social-media baseline.

Why KEV membership is reported but not used as the training label. Even with only ~ 22 KEV CVEs in Megavul, the column is informative and is preserved in the converted records. It is useful for spot-checking the model’s high-EPSS predictions against catalog ground truth: do the KEV CVEs receive high predicted scores? But it is too sparse to drive optimisation directly. The EPSS-thresholded binary label, while noisier, has roughly seven times more positives and is what the soft-label runs report against.

2.5.2 Megavul Variant Map

The 15 Megavul runs in this round use the same five-way ablation pattern as the 20 social-media runs. The summary mapping is different.

Variant	Description	LLM summary used
D	yes	none
S_URL	no	commit-URL summary
S_CODE	no	code summary (vulnerable function)
S_CVSS	no	CVSS-metrics summary
ALL	yes	all three concatenated

Table 8: Megavul variant map. Fifteen runs total: three datasets $\times$ five variants. S_URL and S_CODE replace the S_ALL and S_GIT labels used on the social-media side, because those columns do not exist in the Megavul schema.

2.6 NVD/KEV Dataset

The NVD/KEV dataset is built directly from NVD records, the CISA KEV catalog, FIRST EPSS scores, and the ExploitDB snapshot. Unlike the LLM-summary corpora, it does not come from a third-party CSV; the data-collection step assembles it.

File family	Records	KEV positives	Positive rate
Full labelled records	132,322	803	0.61%
Balanced labelled records (v1)	4,015	803	20.00%
Balanced labelled records (v2)	4,015	803	20.00%

Table 9: NVD/KEV dataset sizes.

The full file covers CVEs published from 2020 through 2024. The balanced files keep all 803 KEV positives and sample 3,212 non-KEV negatives. The earlier multiview-hybrid runs used the second balanced file (the v2 file).

KEV catalog evolution. The 803 figure is fixed at the time the labels were generated; the local CISA KEV snapshot at that time held 1,554 entries. The current authoritative CISA KEV catalog has 1,590 CVEs (catalogVersion 2026.05.08, released 2026-05-08). Re-binding the labels to the current KEV would require rerunning the data-collection step and rebuilding the affected experiments; we have not done so for this round. The KEV cross-reference reported alongside

model predictions in the experiment results document uses the current 1,590-CVE catalog so that the model’s high-confidence non-KEV predictions can be checked against the up-to-date catalog.

Source	Purpose
NVD records dump	Raw NVD records, flattened to description, publication date, CVSS and CWE.
CISA KEV catalog snapshot	The binary label source.
FIRST EPSS dump	EPSS scores and (in the bulk path) percentiles.
ExploitDB snapshot	Public exploit and PoC metadata used by the v2 balanced file.
Merged labelled records	Final per-CVE training records consumed by the dataset module.

Table 10: Sources used to build the NVD/KEV dataset.

2.6.1 NVD/KEV Label Generation

The binary label is dataset-dependent. For the NVD/KEV file it is generated from CISA KEV membership:

$$\text{binary_label}_{\text{KEV}} = \begin{cases} 1, & \text{if CVE-ID is in CISA KEV} \\ 0, & \text{otherwise.} \end{cases}$$

The same membership signal is also stored as a separate *in-KEV* flag. For the records we have, the binary label and the in-KEV flag agree on every record, so there is no label conflict.

For the LLM-summary datasets the same binary-label field exists but is generated differently. Those datasets carry no KEV signal, so the adapter thresholds the EPSS score at 0.10:

$$\text{binary_label}_{\text{EPSS}} = \begin{cases} 1, & \text{if EPSS} \geq 0.10 \\ 0, & \text{otherwise.} \end{cases}$$

The text-source ablations do not actually train against this binary label. They use the soft-label mode with the raw EPSS score as a continuous target in $[0, 1]$. The thresholded binary label is only used for stratified splitting and for reporting precision, recall and F1 at a 0.5 decision threshold. The two definitions of binary label should not be conflated when comparing NVD/KEV runs against LLM-summary runs.

2.6.2 NVD/KEV Feature Fields

Table 11: Fields in the NVD/KEV labelled dataset.

Field	Source	Meaning
CVE identifier	NVD	CVE-ID, used as the sample key.
Description	NVD	Main vulnerability text used to build the TPG.
Published timestamp	NVD	CVE publication timestamp; used for the tabular age feature.
CVSS v3 score	NVD	Normalised in the tabular branch. Missing for 0.25% of the full file.

Field	Source	Meaning
CVSS v3 vector	NVD	Vector string parsed into 22 one-hot tabular dimensions.
CWE list	NVD	CWE identifiers, encoded as top-K multi-hot features in hybrid mode. Empty for 11.40% of the full file.
References	NVD	Reference URLs. Present in the collector schema, but absent from the saved files.
Binary label	CISA KEV	Target for binary-label runs. 1 means the CVE is in KEV.
In-KEV flag	CISA KEV	Boolean copy of the same KEV membership signal.
EPSS score	FIRST EPSS	EPSS probability. In binary KEV experiments this is an optional input feature, not the label.
EPSS percentile	FIRST EPSS	EPSS percentile. Present in the v2 balanced file; absent from the older full file.
High-EPSS flag	derived from EPSS	Convenience flag using a 0.5 EPSS cutoff.
Public-exploit flag	ExploitDB	Whether public PoC or exploit code was found. Present in the v2 balanced file.
Number of exploits	ExploitDB	Number of mapped public exploit records. v2 only.
Verified-exploit flag	ExploitDB	Whether at least one mapped ExploitDB entry is verified. v2 only.
Exploit types	ExploitDB	Exploit category list (remote, webapps, local, DoS, etc.). v2 only.
Earliest exploit date	ExploitDB	Earliest public exploit date when available. v2 only.

2.6.3 Pipeline Use

The NVD/KEV dataset is the default training input when no `--source-csv` is provided. Two common modes:

- Without `--skip-collect`, the pipeline fetches NVD, KEV, EPSS and ExploitDB data, then writes the merged labelled records.
- With `--skip-collect`, the pipeline loads existing labelled records, either from the default location or from a path passed via `--labeled-file`.

The LLM-summary datasets are read instead via the `--source-csv` flag. The dataset adapter maps the relevant raw columns to the labelled-records format, picks the chosen summary column based on `--summary-source`, and writes per-run labelled records.

2.6.4 Robustness Check

The saved NVD/KEV records pass the basic consistency checks:

- every positive label in the full file is present in the KEV catalog snapshot;
- every KEV CVE present in the full NVD file is labelled positive;
- the binary label and the in-KEV flag agree on every record;

- the balanced files are subsets of the full labelled file;
- the v2 balanced file contains all 803 positives;
- descriptions are non-empty after rejected CVEs are removed;
- EPSS scores are in the valid $[0, 1]$ range.

A few caveats remain:

- The balanced files are saved artefacts. The exact negative-sampling step is not part of the main pipeline, so these files should be treated as fixed unless a new reproducible sampler is added.
- KEV is conservative. A label of 0 means “not in CISA KEV”, not “never exploited”.
- KEV labels reflect the catalog snapshot at collection time. For temporal experiments this matters because a CVE can be added to KEV after its publication date.
- The dataset cache hashes the labelled records and invalidates processed graph caches when the source changes. This matters because older runs reused the same cache root for different labelled inputs.

3 Raw Dataset Columns

The schema below covers the new format used by GPT, Gemma and Mistral. Where DeepSeek differs (older naming), the difference is noted in the “Pipeline use” column.

Table 12: Raw dataset columns and how they are used.

Field	Type	Meaning	Pipeline use
CVE identifier	string	The CVE-ID.	Used as the record key, not as a predictive feature.
Source platform	categorical	Collection source for the row (Telegram, Reddit, Hacker News, ExploitDB, Mastodon, BleepingComputer, etc.).	Stored as metadata. Not encoded by the model.
Date posted	date string	Observation date from the source collection. (DeepSeek uses a renamed date field.)	Combined with the time field to form the publication timestamp. Used by the tabular age feature.
Time posted	time string	Time component of the observation. (DeepSeek uses a renamed time field.)	Combined with the date field to form the publication timestamp.
Social-media post	string	Original social or source text mentioning the CVE. (DeepSeek uses a renamed text field.)	Not used by the TPG pipeline. The TPG text comes from the description and optionally from an LLM summary.

Field	Type	Meaning	Pipeline use
EPSS score	float	FIRST EPSS exploitation probability.	Used as the soft label in soft-label mode. Removed from the input by <code>--no-epss-feature</code> to prevent target leakage.
EPSS status	categorical	Marks whether the EPSS value is original or enriched.	Not used by the model. Useful for dataset auditing.
Description	string	NVD vulnerability description.	The main TPG text. Excluded only by <code>--summary-only-tpg</code> .
CVSS version	numeric	CVSS version (3.0, 3.1).	Not encoded directly. The adapter rebuilds the CVSS vector from the score and component columns.
CVSS score	float	CVSS base score, 0 to 10.	Normalised in the tabular branch.
Attack Vector	categorical	CVSS attack vector (NETWORK, LOCAL, ADJACENT_NETWORK, PHYSICAL).	Rebuilt into the CVSS vector string and one-hot encoded.
Attack Complexity	categorical	CVSS attack complexity (LOW, HIGH).	Rebuilt and one-hot encoded.
Privileges Required	categorical	Privileges required (NONE, LOW, HIGH).	Rebuilt and one-hot encoded.
User Interaction	categorical	User interaction needed (NONE, REQUIRED).	Rebuilt and one-hot encoded.
Scope	categorical	Whether exploitation changes scope (UNCHANGED, CHANGED).	Rebuilt and one-hot encoded.
Confidentiality impact	categorical	Confidentiality impact (NONE, LOW, HIGH).	Rebuilt and one-hot encoded.
Integrity impact	categorical	Integrity impact (NONE, LOW, HIGH).	Rebuilt and one-hot encoded.
Availability impact	categorical	Availability impact (NONE, LOW, HIGH).	Rebuilt and one-hot encoded.
Occurrence count	integer	Number of source or social records associated with the CVE. (DeepSeek uses a renamed source-count field.)	Mapped to the social-source count and the exploit count. Used as a fallback signal-richness feature.
Sources available	boolean	Whether source links or source text were available.	Not encoded by the model.
Code-available flag	boolean/integer	Whether public exploit code or PoC is available on GitHub. (DeepSeek uses a renamed code-available field.)	Mapped to the public-exploit flag. Used as a binary tabular feature.
Days since latest GitHub source	float	Days since the most recent linked GitHub source. (DeepSeek uses a renamed delta-days field.)	Not currently used by the model.
Days since oldest GitHub source	float	Days since the oldest linked GitHub source. (DeepSeek uses a renamed delta-days field.)	Not currently used by the model.

Field	Type	Meaning	Pipeline use
Source links	string/list	URLs or source links used as evidence for the CVE row.	Not encoded directly. Upstream evidence for the summary columns.
All-sources summary	string	LLM summary built from every available source link.	Selected by <code>--summary-source all_sources</code> . Combined into the LLM summary by <code>--summary-source combined</code> .
GitHub URLs	string/list	Raw GitHub URLs for the CVE. (Not present in DeepSeek.)	Upstream evidence for the GitHub-URLs summary. Not encoded directly.
GitHub-URLs summary	sum-string	LLM summary built from GitHub URLs only.	Selected by <code>--summary-source github_urls</code> . Combined by <code>--summary-source combined</code> .
CVSS-metrics summary	sum-string	LLM summary written from the CVSS metrics in templated form. (Not present in DeepSeek.)	Selected by <code>--summary-source cvss_metrics</code> . Combined by <code>--summary-source combined</code> .

4 Converted Fields in the Labelled Records

Table 13: Fields written by the dataset adapter and consumed by the TPG dataset.

Converted field	Source field(s)	Meaning	Used by model
CVE identifier	raw CVE field	Sample identifier.	Metadata only.
Description	raw description	NVD text used to build the TPG.	Yes, except in <code>--summary-only-tpg</code> mode.
Publication timestamp	date + time fields	Timestamp proxy.	Yes; used for the tabular age feature.
CVSS v3 score	raw CVSS score	CVSS v3 score.	Yes, when the hybrid head is enabled.
CVSS v3 vector	CVSS component columns	Reconstructed CVSS vector string.	Yes; parsed into the 22 one-hot dims.
CWE list	not provided	CWE list. Empty in these datasets.	Encoded as zeros (CWE features).
References	not provided	NVD reference list. Empty.	Falls back to the source count.

Converted field	Source field(s)	Meaning	Used by model
Binary label	raw EPSS thresholded at 0.10	Binary high-risk label.	Used for stratified splits and threshold-0.5 metrics, not as the training target.
EPSS score	raw EPSS	Soft EPSS target.	Label in soft-label runs.
EPSS percentile	default 0.0	Percentile is unavailable here.	Removed by <code>--no-epss-feature</code> .
High-EPSS flag	raw EPSS ≥ 0.10	High-EPSS convenience flag.	Not encoded directly.
In-KEV flag	default false	KEV status unavailable in these datasets.	Not used here.
Public-exploit flag	raw code-available field	Whether public exploit or PoC code is available.	Yes; binary tabular feature.
Exploit count	raw occurrence count	Count proxy.	Yes; log-normalised.
Social-source count	raw occurrence count	Same count, kept as a separate field.	Yes; fallback for the references count.
Code-available flag	raw code-available field	Boolean, kept as a separate field.	Used as a fallback.
LLM summary	one of the three summary columns, or all three joined	Unified summary text.	Yes, when summary-related TPG flags are set.
Source platform	raw source field	Original source platform.	Metadata only.

5 TPG Features

The graph branch does not consume raw columns directly. It converts text into a graph whose nodes represent documents, sentences, tokens, entities, predicates, arguments, phrases, clauses and security-specific objects. Edges represent containment, sequence, syntax, semantic relations, coreference, discourse, and (when enabled) security relations.

The edge features in this table are graph features. They live on TPG edges and are consumed by edge-aware GNN backbones during message passing. They are not part of the tabular feature vector. The multi-view TPG formula document covers the per-view encoder, the node-conditioned fusion, and the pooling step that produce the graph embedding.

6 Tabular Features

When `--hybrid` is set, the graph embedding is concatenated with a tabular feature vector $\mathbf{f}_{\text{tab}}$. After the leakage-fix pass, the default configuration is 55-dim: the EPSS score and percentile are removed from the input because the soft target is EPSS itself. The 57-dim variant (with EPSS as input) is no longer the default and is kept only for the NVD/KEV binary experiments, where the target is KEV membership rather than EPSS.

Graph feature	Created from	Dimensions / values
Node-type one-hot	TPG schema (13 base types)	13-dim indicator over DOCUMENT, PARAGRAPH, SENTENCE, TOKEN, ENTITY, PREDICATE, ARGUMENT, CONCEPT, NOUN_PHRASE, VERB_PHRASE, CLAUSE, MENTION, TOPIC.
Text embedding	SecBERT (jackaduma/SecBERT, frozen)	768-dim contextual embedding per node. Nodes without text get a zero vector.
Source one-hot (optional)	<code>--add-source-labels</code>	3-dim indicator: description / summary / mixed.
<code>edge_index</code>	TPG edges	$2 \times M$ integer tensor with directed connectivity. Used by PyG message passing.
<code>edge_type</code>	Edge relations	M -vector of integer relation indices. 13 base types (0...12); 23 total when <code>--include-security-edges</code> adds the SEC relations (13...22).
<code>edge_attr</code>	<code>edge_type</code>	One-hot edge-type vector (13- or 23-dim).
Security edges	<code>--include-security-edges</code>	Adds the SEC relations AFFECTS, HAS_VERSION, LOCATED_IN, CLASSIFIED_AS, EXPLOITED_BY, CAUSES, MITIGATED_BY, USES_FUNCTION, THREATENS, HAS_SEVERITY (10 types).
Source labels	summary / two-view flags	Marks nodes and edges as description, summary or mixed.
Summary pooling node	<code>--summary-pooling-node</code>	Adds one synthetic node linked to the document via CONTAINS, with a pooled summary embedding.

Table 14: Main features used by the TPG branch. The base node feature dimension is $d_{\text{in}} = 13 + 768 = 781$, or 784 with source labels appended.

The tabular vector contains CVE-level metadata: CVSS, CWE, vulnerability age, source counts and public-exploit availability. It does not contain edge types or SEC edge labels; those live in the graph branch.

7 Which Features Are Used by Each Experiment

The social-media text-source ablation uses five mutually exclusive variants. The five rows below match the variants reported in the experiment results.

The Megavul runs use the same variant scheme but a different summary column map; see Section 2.5 for the `S_URL` / `S_CODE` / `S_CVSS` mapping. Targets, tabular branch and EPSS handling are identical to the social-media table above.

The other experiment types use different combinations.

8 Leakage Notes

The main leakage risk in the LLM-summary experiments is using the EPSS score as an input feature while also predicting EPSS. That puts the target directly into the input and inflates the metrics. The leakage-free EPSS soft-label experiments remove this input with `--no-epss-feature`.

Feature group	Dim	Meaning
CVSS score and has-CVSS flag	2	Normalised CVSS v3 score in $[0, 1]$ plus a binary flag indicating whether CVSS data exists.
CVSS vector one-hot	22	One-hot encoding of the eight CVSS v3 metrics (see Table 16).
CWE multi-hot and “other” bucket	26	Top-25 CWE IDs plus a catch-all bucket. Populated for NVD/KEV; all zero for the LLM-summary datasets (no CWE list there).
CWE count	1	Normalised count of CWE IDs (capped at 10, divided by 10). Populated for NVD/KEV; zero for the LLM-summary datasets.
Reference or source count	1	Log-normalised $\log(1 + n)/\log(21)$ of NVD reference URLs, or social/source mention count when references are absent.
Vulnerability age	1	Log-normalised age in days from the publication-date proxy, $\log(1 + \text{age})/\log(3651)$.
EPSS score and percentile (optional)	2	Direct EPSS inputs. Excluded by default. Included only in the NVD/KEV binary runs where EPSS is not the label.
Public-exploit flag	1	Binary indicator that public PoC or exploit code is available.
Exploit/source count	1	Log-normalised count proxy from the occurrence count, with the social-source count as a fallback.

Table 15: Tabular features used by the hybrid model. Default total is $\dim \mathbf{f}_{\text{tab}} = 55$ (rows excluding the optional 2-D EPSS group). With EPSS included it is 57.

In the NVD/KEV binary experiments the EPSS score is not the target; KEV membership is. EPSS is still a strong external exploitation-risk signal, so results with EPSS included should be read as hybrid prediction with an external risk signal, not as a pure text-only result.

A second, more subtle risk is that the CVSS-metrics summary is just templated CVSS prose. The same CVSS information is already in the 22-dim one-hot tabular block, so feeding it again as text adds redundancy and seems to displace the description signal. The text-source ablations show this empirically: `S_cvss` is the worst single-source variant on every dataset.

Other fields are not labels, but some are strong proxies. CVSS severity, public-exploit availability, source count and the LLM summaries all carry information related to exploitation likelihood. They are useful modelling features, but the reported results should make clear which of these signals were available to the model.

CVSS v3.1 metric	Cardinality	Possible values
Attack Vector (AV)	4	NETWORK, ADJACENT_NETWORK, LOCAL, PHYSICAL
Attack Complexity (AC)	2	LOW, HIGH
Privileges Required (PR)	3	NONE, LOW, HIGH
User Interaction (UI)	2	NONE, REQUIRED
Scope (S)	2	UNCHANGED, CHANGED
Confidentiality (C)	3	NONE, LOW, HIGH
Integrity (I)	3	NONE, LOW, HIGH
Availability (A)	3	NONE, LOW, HIGH
Total	22	

Table 16: Decomposition of the 22-dim CVSS one-hot.

Variant	Description	LLM summary used	Tabular	EPSS input	Security edges
D	yes	none	yes	no	no
S_ALL	no	all-sources summary	yes	no	no
S_GIT	no	GitHub-URLs summary	yes	no	no
S_CVSS	no	CVSS-metrics summary (not on DeepSeek)	yes	no	no
ALL	yes	all three concatenated	yes	no	no

Table 17: Feature use across the five social-media text-source variants.

Experiment	Description	LLM summary	Tabular	EPSS input	Security edges
NVD/KEV binary rerun	yes	no	yes	yes (not the label)	no
NVD/KEV temporal shift	yes	no	yes	yes	no
NVD/KEV no-EPSS extrapolation	yes	no	yes	no	no

Table 18: Feature use across the NVD/KEV experiments.